

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA

DIPLOMA EXAMINATION (C-24)

C-24-DEC-2025

SEMESTER III, SEMESTER END EXAM

AI/BM/CCB/CPS/CS/EC/EE/EI/ES/EV

EC-302

Digital Electronics



**PCODE
243034**

Exam Date: 02-12-2025

Session: FN

Duration: 2 Hours [10:00 AM To 12:00
NOON]

[Total Marks: 40]

PART-A

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

8 X 1 = 8

1. State sign bit numbers for positive, negative numbers in binary system.
2. List applications of encoder
3. What is a Register?
4. Define a combinational logic circuit.
5. Draw R-S Flip-Flop using NAND gates.
6. List the different types of shift Registers.
7. What is volatile memory?
8. list any two commonly used IC numbers of counters

PART-B

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

4 X 3 = 12

- 9(a). Explain the basic logic gates AND, OR gates with truth table.
--- OR ---
- 9(b). List the different types of Triggering in Flip flops
- 10(a). Write the truth table of Half adder circuit.
--- OR ---
- 10(b). How many flip flops are required to design divide by 64 counter. Give reasons.
- 11(a). Draw the circuit diagram of bi-directional shift register.
--- OR ---

11(b). What is race around condition in JK flip-flop and give methods to avoid it? ✓

12(a). What is a ring counter? Draw its diagram. ✓

--- OR ---

12(b). Draw the Circuit diagram of Diode ROM .

PART-C

Instructions:

1. Answer the following questions.

4 X 5 = 20

2. Each question carries **FIVE** marks.

13(a). Realize EX-NOR Gate by using NAND & NOR gates. ✓

--- OR ---

13(b). Explain RS latch using NAND gates only and NOR gates only with truth tables

14(a). Draw the circuit diagrams and truth tables of 4X1 multiplexer and 1X4 demultiplexer along with truth tables and IC numbers

--- OR ---

14(b). Explain the working of 4-bit synchronous counter with a circuit and timing diagram. ✓

15(a). Explain the working of 4-bit shift left register with a circuit and timing diagram. ✓

--- OR ---

15(b). Explain the operation of T flip flop using JK flip flop and draw the truth table and timing diagram.

16(a). Distinguish between SRAM and DRAM.

--- OR ---

16(b). Distinguish between EEPROM and UVPROM. ✓